QuantAnomalies

Quantitative Anomalies as Market Opportunities

Progress Presentation • Statistical & AI Dashboard for Equity and Forex Markets

1 Title of the Study

Quantitative Anomalies as Market Opportunities — A Statistical & AI Dashboard for Equity and Forex Markets.

> In plain terms: textbook finance says markets are "efficient" — prices already reflect everything, risk never changes, and nothing is predictable. Reality is messier, and those cracks in the theory are recurring patterns called **anomalies**. Our project hunts for those anomalies and reframes them as **opportunities**: places where the odds may tilt in a careful observer's favour.

2 Objectives of the Study

Each objective targets one well-known market anomaly and turns it into a usable opportunity signal.

1. **Macro-driver opportunity.** Measure *which* macroeconomic forces (interest rates, inflation, the fear index, oil, gold, the dollar) move an asset's returns — and crucially with *what time delay*. If a driver acts weeks later, that lag is a window to anticipate moves.

2. **Risk-regime opportunity.** Model how risk itself rises and falls over time (volatility clustering) and detect that crashes happen far more often than a bell curve predicts (fat tails). Spotting a "stormy" regime early is a chance to cut risk — or to recognise unusually cheap calm.

3. **Mean-reversion opportunity.** Find pairs of currencies whose prices are tethered over the long run (cointegration), then flag when they drift unusually far apart — history says the gap tends to snap back. This is the basis of market-neutral "statistical arbitrage".

4. **Opportunity synthesis (the recommendation engine).** Combine all of the above into one automated scanner that ranks what is unusual or actionable *right now*, and uses a **local AI analyst** to explain each finding in plain English.

5. **Accessibility & robustness.** Make it work for *any* ticker the user types, on a reliable data pipeline, presented through a clear, interactive dashboard.

3 Motivation for Selecting the Topic

- **The anomalies are real and well-studied, but rarely made tangible.** Lagged macro effects, volatility clustering, fat tails, and cointegration all appear in finance literature; we wanted to make them *visible and actionable* rather than abstract equations.

- **Reframing risk as opportunity.** The same statistics used to warn about risk can also point to opportunity — we wanted a tool that does both.

- **Bridging statistics and modern AI.** We combine classic econometrics (regression, GARCH, cointegration) with a **local large language model** that turns numbers into a readable analyst note — private, free to run, and on our own GPU.

- **Learning value.** The project spans real statistical modelling, live financial-data engineering, full-stack web development, and on-device AI.

4 Data & APIs Used

In place of a questionnaire and a survey sample: our "instrument" is a set of financial-data APIs, and our "sample" is the live market history they return.

A. External data-source APIs (collecting raw market data)

- **Yahoo Finance** (via the `yfinance` library) — daily price history (2015–2025) for any equity, index, currency pair, future, or crypto. Examples: `^GSPC` (S&P 500), `^VIX` (volatility index), `CL=F` (oil), `GC=F` (gold), `^TNX` (10-year Treasury yield), `DX-Y.NYB` (US dollar index), and 45 forex pairs.
- **FRED — U.S. Federal Reserve Economic Data** — macroeconomic series: CPI (inflation), Fed Funds Rate, and Unemployment. A free mirror, **DBnomics**, is a built-in backup so a single outage never stops the study.

B. Our own analysis API (FastAPI backend)

Turns raw data into results:

- `/api/macro-regression/*` — regression, Granger-causality, correlation, time series
- `/api/garch/*` — volatility model, clustering tests, return distribution
- `/api/pairs/*` — cointegration tests, best-pair spread & signals, correlation
- `/api/recommendations` — the ranked opportunity scan
- `/api/llm/info` — status of the local AI analyst

C. Local AI API

An on-device `llama.cpp` server (OpenAI-compatible) running the **Qwen3-4B** model on our **NVIDIA RTX 4050 GPU**, used to explain detected opportunities in plain language. No data leaves the machine.

5 Pilot Study Conducted to Date

We have a **working end-to-end prototype**, validated on real data.

Data pipeline (validated). Live downloads work for all sources; a corrupted early cache (the volatility and oil series had been mixed up) was found and fixed. Every series now sits in its correct real-world range — e.g. VIX 9–83, oil -37 to 124, 10-year yield 0.5–4.9%.

Three analysis pillars — working, with sample findings on the S&P 500:

- *Macro:* macro factors explain about **64%** of monthly return variation ($R^2 \approx 0.64$); the model surfaces which factors are significant and at what lag.
- *Volatility:* GARCH "persistence" ≈ **0.9955** (shocks fade very slowly) and excess kurtosis ≈ **16** (strong fat tails) — risk is *not* constant and crashes are under-estimated by the normal curve.

- *Pairs:* among major forex pairs, **USDCHF/USDJPY** tested cointegrated ($p \approx 0.008$) with a mean-reversion half-life around 64 days.

Recommendation engine (v1 working). Four detectors (volatility regime, tail move, trend, forex mean-reversion) rank opportunities by severity and assign a confidence score. The local AI produces a grounded plain-English note in ≈4 seconds on the GPU (≈28 tokens/sec), citing only the real detected numbers.

Dashboard (working). Interactive web app: search any ticker, pick forex pairs, switch time ranges, and read explained results across five tabs (Overview, Opportunities, Macro, GARCH, Pair Trading).

> **In short:** the pilot confirms the data, the statistics, the AI-explanation layer, and the user interface all work together on live markets. Next steps (standardised comparisons, an options / Black–Scholes module, and a back-test of the signals) are tracked in the project roadmap.

Appendix — Each anomaly in one line (layman)

- **Lagged macro effect:** *News ripples through markets over weeks, not instantly — know the lever and the delay, and you can position ahead.*
- **Volatility clustering:** *Calm days cluster, and so do wild days — storms arrive in groups, giving you warning to manage risk.*
- **Fat tails:** *Extreme crashes happen much more often than the textbook bell curve says — plan for them.*
- **Cointegration / pairs:** *Two currencies on one leash: when they drift far apart, they usually snap back — bet on the gap closing, up market or down.*